

Vanshika Agrawal

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EDUCATION

New York University (NYU)

Master of Science in Computer Engineering (GPA: 3.8/4.0)

New York, NY

Aug 2025 - May 2027

- Relevant Coursework: Efficient AI, Machine Learning, Data Engineering, ML Systems, Big Data

Bhilai Institute of Technology (CSVТУ)

Bachelor of Technology in Electronics & Telecommunications (GPA: 9.5/10.0)

Durg, IN

Aug 2018 - May 2022

TECHNICAL SKILLS

Languages & Core Frameworks: Python, C, C++, Java, SQL, PyTorch, TensorFlow, Sklearn, Pandas, NumPy, LangChain, HuggingFace

AI/ML Infrastructure & Modeling: LLMs, RAG, Agentic Workflows, Computer Vision, Graph Generative Models, Deep Learning, Vision-Language Models (VLMs), Model Quantization, Multimodal Grounding, Evaluation Harnesses, Time-Series Forecasting

Data & Distributed Systems: Docker, Kubernetes, Hadoop, Airflow, Kafka, Spark, MLflow, CI/CD, Git, ETL, Serving, Prometheus

Cloud and Databases: AWS (S3, EC2, Athena, SageMaker), GCP (BigQuery, Vertex AI), Terraform, MySQL, Vector Databases

Honors & Achievements: 2nd Place - Datadog Agentic Hackathon (Solo), Outreach Chair ACM-W, CSC Graduate Co-Representative

WORK EXPERIENCE

Enkira AI

New York, NY

AI Engineer Intern

June 2026 - Aug 2026

- Architected a modular multi-agent UI perception framework utilizing specialized sub-agents (Discovery, Vision, Fidelity) to parse complex structures, discovering and sequencing 13+ dynamic tabbed interfaces via accessibility role signals.
- Profiled generative asset omissions via controlled regression experiments; deployed a vision-language verification pipeline utilizing a Rust-based DevTools CLI to capture hidden UI features, mitigating silent data loss across 115+ structural entries.

New York University

New York, NY

Research Assistant

June 2026 - Aug 2026

- Investigated structural mode collapse in graph generative models across GANs, VGAEs, and Diffusion Models using PyTorch, establishing that 90%+ of synthetic graphs exhibit geometric distortions across Forman and Ollivier-Ricci curvatures.
- Synthesized curvature-based geometric evaluation pipelines applying Leiden community detection across 270 generated graphs, capturing a 0.31 ARI / 0.40 NMI standout alignment to quantify structural fidelity loss in latent embedding spaces.

Accenture

Mumbai, IN

Software Engineer

Dec 2022 - July 2024

- Scaled ML anomaly detection systems across 50K+ industrial assets using PyTorch, TensorFlow, and Scikit-learn with MLflow experiment tracking and targeted feature engineering to cut inspection costs by 23% and lower the false-positive rate.
- Architected end-to-end ETL and data orchestration pipelines using Python, Pandas, and NumPy across 178K+ asset records with automated SQL validation and lifecycle management, eliminating 3,500+ manual engineering hours annually at 97.9% uptime.

JPMorgan Chase

Mumbai, IN

Site Reliability Engineer - Risk Management (FX, Commodities, Emerging Markets)

Jul 2022 - Nov 2022

- Orchestrated high-throughput, fault-tolerant data streaming pipelines via Kafka and Spark Streaming to ingest 10GB+ of daily telemetry metrics, serving real-time feature streams to low-latency analytics dashboards and 8+ live trading systems.
- Optimized hybrid-cloud production ETL workflows across Airflow, Spark, AWS Athena, and GCP BigQuery, profiling distributed compute bottlenecks to slash critical system incident latency by 42.9% and accelerate data orchestration runtime.

PROJECTS

Synapse Riders - On-Device Memory Recall Assistant for Cognitive Decline

Top 30 Finalist - ExecuTorch Hackathon (Meta, Qualcomm, Samsung)

- Engineered a multimodal, 100% offline edge AI system on the Snapdragon 8 Elite NPU, cross-compiling models across Meta's ExecuTorch, Qualcomm QNN, and QAI RT Genie runtimes to support local face, voice, and LLM inference.
- Designed a localized RAG knowledge graph via Android Room SQL, pairing 512-dimensional ArcFace cosine similarity with geometric center-crop targeting to enable low-overhead identity matching and grounded query synthesis under 200ms.

SuppGo - Agentic LLM Hallucination Detection and Audit System

Second Prize Winner - NYU PULSE Hackathon

- Authored an agentic hallucination detection engine via LangChain and Python utilizing Hugging Face embeddings, multi-step RAG reasoning, and query decomposition across 100+ domain-specific queries to maximize factual grounding accuracy.
- Instrumented multi-stage NLP pipeline execution passes spanning web ingestion loops, structured extraction layers, vector retrieval, and external knowledge-base cross-validation, enforcing factual grounding constraints to outperform 40+ teams.

Hybrid LSTM-GARCH Volatility Forecasting for Options Pricing

- Trained a hybrid LSTM-GARCH deep learning pipeline in PyTorch for financial time-series volatility forecasting, combining neural sequence modeling with econometric structures to reduce predictive RMSE by 15%+ over baseline ARIMA models.
- Containerized the model inference engine via Docker to automate multi-asset backtesting, executing Diebold-Mariano and Wilcoxon signed-rank tests across 3+ datasets to establish statistical significance at $p < 0.05$ for algorithmic options pricing.